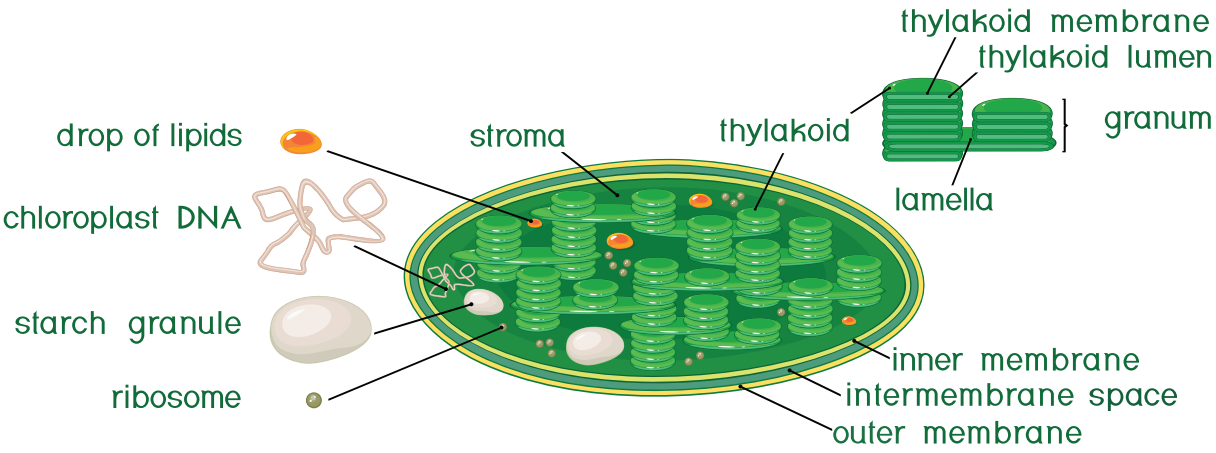


Figure 6.19: Structure of Chloroplast

6.6.7 Ribosome

Ribosomes were first observed by **George Palade** (1953) as dense particles or granules in the electron microscope. Electron microscopic observation reveals that ribosomes are composed of two rounded sub units, united together to form a complete unit. Mg^{2+} is required for structural cohesion of ribosomes. Biogenesis of ribosome are *denova* formation, auto replication and nucleolar origin. Each ribosome is made up of one small and one large sub-unit Ribosomes are the sites of protein synthesis in the cell. Ribosome is not a membrane bound organelle (Figure 6.20).

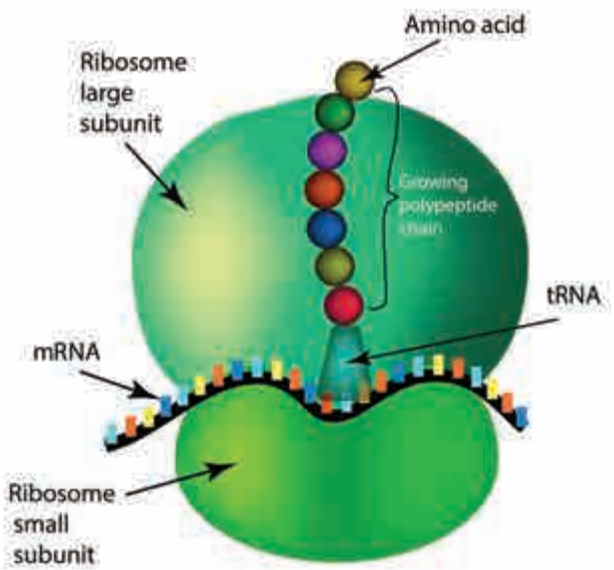


Figure 6.20: Structure of Ribosomes

Types of Ribosomes

70S Ribosomes (sub unit 30S and 50S)	80S Ribosomes (sub units 40S and 60S)
3 RNA molecule (i) 16SrRNA in 30S subunit (ii) 23S and 5S in 50S large subunit **(Prokaryotic cells of Bluegreen algae Bacteria, Mitochondria and Chloroplast of many Algae and higher plants)**	4 RNA molecule (i) 18SrRNA in 40S small subunit (ii) 28S, 5.8S and 5S in larger 60S subunit **(Eukaryotic cells of Plants and animals)**

DO YOU KNOW?

Svedberg unit (s).

The size of ribosomes and their subunits are usually given in Svedberg unit (named after Theoder Svedberg, Swedish Chemist Noble Laureate 1929), a measure of a particle size dependent on the speed with which particle sediment in the ultracentrifuge.